

Quiz (Habanero)

Q1. Solve for x : $x^2 + 5x + 6 = 0$

- (A) $x = -2, -3$
- (B) $x = -1, -6$
- (C) $x = 2, 3$
- (D) $x = -2, 3$
- (E) $x = 1, -6$

Q2. Factor the quadratic expression: $x^2 - 7x + 12$

- (A) $(x - 3)(x - 4)$
- (B) $(x + 3)(x + 4)$
- (C) $(x - 2)(x - 6)$
- (D) $(x + 2)(x + 6)$
- (E) $(x - 1)(x - 12)$

Q3. Solve for y : $y^2 - 9y + 20 = 0$

- (A) $y = 4, 5$
- (B) $y = 2, 10$
- (C) $y = -4, -5$
- (D) $y = 4, -5$
- (E) $y = 2, -10$

Q4. Factor the quadratic expression: $x^2 + 2x - 15$

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 2)(x - 15)$
- (D) $(x - 2)(x + 15)$
- (E) $(x + 1)(x - 15)$

Q5. A water tank can hold x gallons of water. The tank is currently

$\frac{23}{40}$ full. If x satisfies the equation $x^2 - 12x + 36 = 0$, how many gallons of water are in the tank?

- (A) 8 gallons
- (B) 12 gallons
- (C) 16 gallons
- (D) 20 gallons
- (E) 24 gallons

Q6. Solve for z : $z^2 + z - 6 = 0$

- (A) $z = -3, 2$
- (B) $z = -2, 3$
- (C) $z = -1, 6$
- (D) $z = 1, -6$
- (E) $z = -2, -3$

Q7. A forest's temperature changes according to the quadratic equation $t^2 - 4t - 5 = 0$, where t is the time in hours. What are the possible times when the temperature is 0°C ?

- (A) $t = -1, 5$
- (B) $t = 1, 5$
- (C) $t = -1, -5$
- (D) $t = 1, -5$
- (E) $t = 5, -1$

Q8. Factor the quadratic expression: $x^2 - 4x - 21$

- (A) $(x - 7)(x + 3)$
- (B) $(x + 7)(x - 3)$
- (C) $(x + 2)(x - 21)$
- (D) $(x - 2)(x + 21)$
- (E) $(x + 1)(x - 21)$

Open Ended Questions (FRQ)

Q9. The depth of water in a lake is modeled by the quadratic equation $d^2 - 6d - 8 = 0$, where d is the depth in meters. Solve for d and verify your solutions.

Q10. A farmer's crop yield is affected by the temperature, which is modeled by the quadratic equation $t^2 - 5t - 14 = 0$, where t is the temperature in degrees Celsius. Solve for t and verify your solutions, considering the context of the problem.

Chef's Tips

- Tip 1: Ensure students understand factoring and quadratic equation solving.
- Tip 2: Remind students to verify their solutions.
- Tip 3: Provide real-world examples to illustrate the applications of quadratic equations.